



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

2004

**PRELIMINARY
HIGHER SCHOOL CERTIFICATE
EXAMINATION**

Mathematics

General Instructions

- Reading time – 5 minutes.
- Working time – 2 periods.
- Write using black or blue pen.
- Board approved calculators may be used.
- All *necessary* working should be shown in every question if full marks are to be awarded.
- Marks may **NOT** be awarded for messy or badly arranged work.
- Start each **NEW** section in a separate answer booklet.

Total Marks - 100 Marks

- Attempt Questions 1 - 5
- All questions are of *equal* value.

Examiner: C. Kourtesis

Question 1

Marks

- (a) Express $\frac{2\pi}{9}$ radians in degrees 1
- (b) Solve $|4 - x| = 1$ 1
- (c) Factorize $3a^2 - 6a$ 1
- (d) Solve the inequality $5 + 3x < 7$ 1
- (e) Simplify $49^{\frac{1}{2}} \times 81^{\frac{1}{4}}$ 1
- (f) Simplify $\log_6 216$ 1
- (g) Express with rational denominator $\frac{1}{\sqrt{6} + 2}$ 2
- (h) On the same set of axes sketch the graphs of $y = \log_{10} x$ and $y = 10^x$ 2
- (i) Factorize $m^2 - 16$ 1
- (j) Simplify $6 - a - 2(3 - 2a)$ 2
- (k) Find the area of the triangle ABC 2
-
- (l) If $f(\theta) = \tan 3\theta + \sin 3\theta$ find $f\left(\frac{\pi}{3}\right)$ in exact form 2
- (m) Factorize $x^2 - 5x + 6$ 1
- (n) Simplify $t + \frac{2t - 1}{3}$ 2

Question 2

Marks

(a) The point $(3k, 2)$ lies on the line $2x - 3y = 10$
Find the value of k 2

(b) Solve the equation $\tan 2\theta = 1$ for $0 \leq \theta \leq 2\pi$ 2

(c) Evaluate $\frac{2+\pi}{2-\pi}$ to 1 decimal place 1

(d) Find the derivative of

(i) $4x^9 - 6x + 1$ 1

(ii) $(5x + 6)^{10}$ 1

(iii) $\frac{x^3 + 10}{x^2}$ 2

(e) Use the quotient rule to differentiate 2

$$y = \frac{x}{3x + 2}$$

(f) Find integers a and b such that 2

$$(5 - \sqrt{2})^2 = a + b\sqrt{2}$$

(g) For the parabola $(x - 2)^2 = 4y$ find the coordinates of the

(i) vertex 1

(ii) focus 1

(h) Write down the roots of the equation 1

$$(1 - 3x)(x + 10) = 0$$

(i) Sketch the following graphs on separate number plane graphs, indicating any intercepts with the coordinate axes 4

(i) $y = x + 1$ (iii) $y^2 = -x$

(ii) $y = \frac{4}{x}$ (iv) $y = 2(x + 3)^2$

Question 3

Marks

- (a) Given that $\log a = 0.86$ and $\log b = 0.42$ evaluate
- (i) $\log\left(\frac{a}{b}\right)$ (ii) $\log\sqrt{ab}$ 3
- (b) If α and β are the roots of $3x^2 - 2x + 1 = 0$
Evaluate:
- (i) $\alpha + \beta$ 1
- (ii) $\alpha\beta$ 1
- (iii) $\frac{1}{\alpha^2\beta} + \frac{1}{\beta^2\alpha}$ 2
- (c) The roots of the equation $mx^2 - x + n = 0$ are -2 and 3 .
Find the values of m and n . 2
- (d) Find $\lim_{n \rightarrow 10} \frac{n^2 - 100}{n - 10}$ 1
- (e) Solve $|3x + 1| > 8$ 2
- (f) If $\sqrt[n]{m} = n^3$ express x in terms of m and n 2
- (g) The lines $4y = bx + 3$ and $y + 2x = 1$ are parallel. Find b . 2
- (h) Solve the inequality $\theta^2 - 4\theta < 0$ 2
- (i) Find the equation of the tangent to the curve
 $y = x^2 - 2x$ at the point where $x = 1$ 2

Question 4

Marks

- (a) If $3x^2 + 4x + 5 \equiv A(x+1)^2 + B(x+1) + C$ find the value of the constants A , B and C . 2
- (b) Given $P(x) = k - 3kx - x^2$ find the values of k for which $P(x) < 0$ for all x . 2
- (c) Solve the equation $x^4 - 10x^2 + 9 = 0$ 2
- (d) The vertex of the parabola $y = x^2 + bx + c$ is at the point $(2, 3)$. Find the values of b and c . 2
- (e) The locus of a point P is described by the equation $x^2 + y^2 - 4x + 2y = 0$ 2
- Find the centre and radius.
- (f) (i) Sketch the graph of $y = x^2 - 6$ and label all intercepts with the axes. 1
- (ii) On the same set of axes carefully sketch the graph of $y = |x|$ 1
- (iii) Find the x coordinates of the two points where the graphs intersect 1
- (iv) Hence solve the inequality $x^2 - 6 \leq |x|$ 1
- (g) Sketch the region in the number plane which satisfies both the inequalities $y \leq \sqrt{16 - x^2}$ and $x - y \geq 4$3
- (h) If $\tan A + \sec A = 2$, evaluate $\cos A$...3

Question 5

Marks

- (a) Use the product rule to differentiate

$$(x^2 + 5)(1 - 4x)$$

3

- (b) Given the function $f(x) = \frac{1}{\sqrt{3-x^2}}$ find the:

(i) domain

2

(ii) range

- (c) Show that $h(x) = x^3 - x$ is an odd function

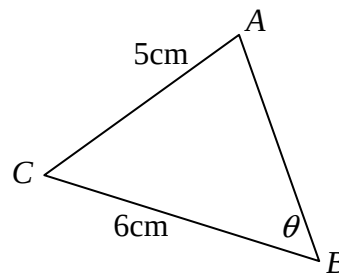
2

- (d) For the triangle ABC

3

Show that

$$\frac{\sqrt{11}}{6} < \cos \theta < 1$$



- (e) $\triangle ABC$ is right-angled at B and DE is perpendicular to AC (see diagram)

- (i) Prove that $\triangle ABC$ and $\triangle CDE$ are similar

2

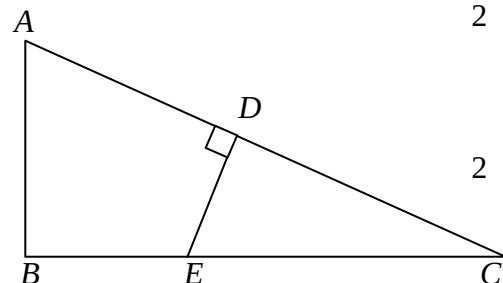
- (ii) Prove that $BC \times CE = AC \times CD$

2

- (iii) Prove that:

$$DE^2 = AD \times DC - BE \times EC$$

2



- (f) If α and β are the roots of $x^2 + 5x + 7 = 0$, form the equation whose roots are $(\alpha - \beta)^2$ and $(\alpha + \beta)^2$

...4



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Mathematics

Sample Solutions

Question	Marker
1	Mr Fuller
2	Mr Boros
3	Mr Dowdell
4	Mr Dunne
5	Ms Nesbitt

Question 1:

$$(a) \frac{2\pi}{9} \times \frac{180}{\pi} = 40^\circ$$

$$(b) |4 - x| = 1$$

$$4 - x = 1 \quad \text{or} \quad -(4 - x) = 1$$

$$-x = -3 \quad -4 + x = 1$$

$$x = 3 \quad x = 5$$

$$(c) 3a^2 - 6a = 3a(a - 2)$$

$$(d) 5 + 3x < 7$$

$$3x < 2$$

$$x < \frac{2}{3}$$

$$(e) 49^{\frac{1}{2}} \times 81^{\frac{1}{4}} = 7 \times 3$$

$$= 21$$

$$(f) \log_6 216 = \frac{\ln 216}{\ln 6} \quad \text{or} \quad \text{let } \log_6 216 = x$$

$$= 3$$

$$216 = 6^x$$

$$6^3 = 6^x$$

$$\therefore \log_6 216 = 3$$

$$(i) m^2 - 16 = (m - 4)(m + 4)$$

$$(j) 6 - a - 2(3 - 2a) = 6 - a - 6 + 4a$$

$$= 3a$$

$$(g) \frac{1}{\sqrt{6} + 2} \times \frac{\sqrt{6} - 2}{\sqrt{6} - 2} = \frac{\sqrt{6} - 2}{6 - 4}$$

$$= \frac{\sqrt{6} - 2}{2}$$

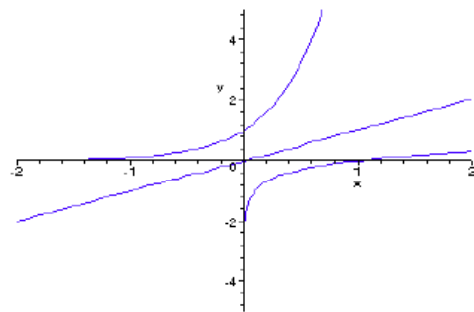
(h)

NB The line $y = x$ is a line of symmetry for the two graphs.

The y axis is a vertical asymptote for the log graph. The x axis is a horizontal asymptote for the exponential graph.

(1,10) would lie on the exponential graph and

(10,1) would be on the log graph.



$$(k) \angle BCA = 30^\circ \quad (\angle \text{ sum of a } \Delta = 180^\circ)$$

$$\text{Area} = \frac{1}{2} \times 6 \times 10 \times \sin 30^\circ$$

$$= 15 \text{ square centimetres}$$

$$(l) \ f(\theta) = \tan 3\theta + \sin 3\theta$$

$$\begin{aligned} f\left(\frac{\pi}{3}\right) &= \tan 3\left(\frac{\pi}{3}\right) + \sin 3\left(\frac{\pi}{3}\right) \\ &= \tan \pi + \sin \pi \\ &= 0 \end{aligned}$$

$$(m) \ x^2 - 5x + 6 = (x - 3)(x - 2)$$

$$\begin{aligned} (n) \ t + \frac{2t-1}{3} &= \frac{3t+2t-1}{3} \\ &= \frac{5t-1}{3} \end{aligned}$$

Q. 2.

(a) $2 \times 3k - 3 \times 2 = 10$

$$6k - 6 = 10$$

$$6k = 16$$

$$k = \frac{16}{6} = \frac{8}{3} = 2\frac{2}{3} \quad (2)$$

(b) $\tan 2\theta = 1$ 

$$2\theta = 45^\circ, 225^\circ, 45+360, 225+360$$

$$2\theta = 45, 225, 405, 585$$

$$\theta = 22\frac{1}{2}^\circ, 112\frac{1}{2}^\circ, 202\frac{1}{2}^\circ, 292\frac{1}{2}^\circ$$

in radians $\frac{\pi}{8}, \frac{5\pi}{8}, \frac{9\pi}{8}, \frac{13\pi}{8} \quad (2)$

(c) $-4.5 \quad (1)$

(d) (i) $36x^8 - 6 \quad (1)$

(ii) $10(5x+6)^9 \times 5 = 50(5x+6)^9 \quad (1)$

(iii) $\frac{x^3}{x^2} + \frac{10}{x^2} = x + 10x^{-2}$

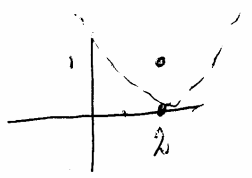
$$y' = 1 + 10x^{-2} \times -2x^{-3}$$

$$= 1 - \frac{20}{x^3} \quad \text{or} \quad \frac{x^3 - 20}{x^3} \quad (2)$$

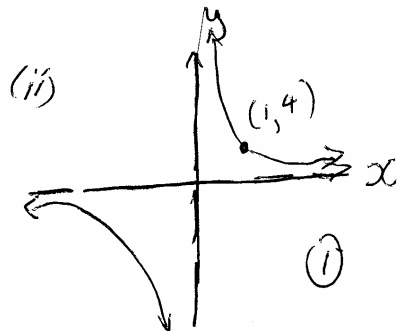
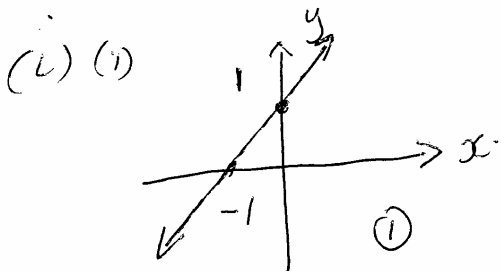
(e) $y' = \frac{(3x+2) \times 1 - x \times 3}{(3x+2)^2} = \frac{3x+2-3x}{(3x+2)^2} = \frac{2}{(3x+2)^2} \quad (2)$

(f) LHS/ $25 - 10\sqrt{2} + 2 = 27 - 10\sqrt{2}$
So $a = 27 \quad (1)$
 $b = -10 \quad (1)$

(3) $(x-2)^2 = 4(y-0)$
 matches with $(x-h)^2 = 4a(y-k)$
 $V(2,0)$ ①
 $a=1$
 $F(2,1)$ ①

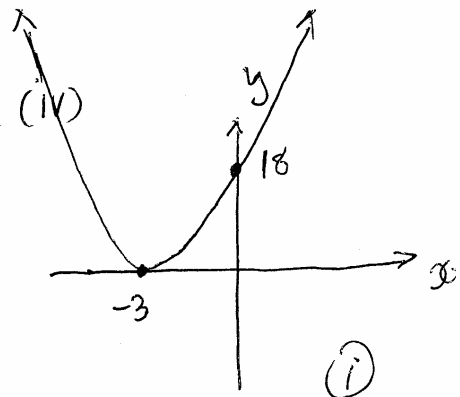
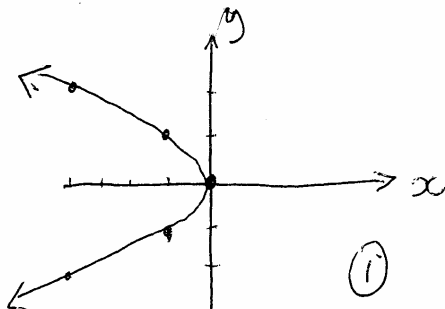


(h) let $1-3x=0$ let $x+10=0$
 $3x=1$
 $x=\frac{1}{3}$ ①
 $x=-10$



(iii) $y^2 = -x$

x	0	-1	-4	-9	-16	-25	-36	-49	-64	-81	-100
y	0	1	2	3	4	5	6	7	8	9	10



Q3

$$(a) (i) \log\left(\frac{a}{b}\right) = \log a - \log b$$

$$= 0.86 - 0.42$$

$$= 0.44 \quad (1)$$

$$(ii) \log \sqrt{ab} = \frac{1}{2} \log(ab)$$

$$= \frac{1}{2} (\log a + \log b)$$

$$= \frac{1}{2} (0.86 + 0.42)$$

$$= 0.64 \quad (2)$$

$$(b) (i) \alpha + \beta = -\frac{-2}{3}$$

$$= \frac{2}{3} \quad (1)$$

$$(ii) \alpha\beta = \frac{1}{3} \quad (1)$$

$$(iii) \frac{1}{\alpha^2\beta} + \frac{1}{\beta^2\alpha} = \frac{\beta + \alpha}{\alpha^2\beta^2}$$

$$= \frac{\left(\frac{2}{3}\right)}{\left(\frac{1}{3}\right)^2}$$

$$= \frac{2}{3} \times \frac{9}{1}$$

$$= 6 \quad (2)$$

$$(c) mx^2 - x + n$$

$$-2 + 3 = \frac{1}{m}$$

$$\therefore 1 = \frac{1}{m}$$

$$\therefore m = 1$$

$$-2 \times 3 = \frac{n}{m}$$

$$\therefore n = -6 \quad (2)$$

$$(d) \lim_{n \rightarrow 10} \frac{(n-10)(n+10)}{n-10}$$

$$= \lim_{n \rightarrow 10} (n+10)$$

$$= 20 \quad (1)$$

$$(e) |3x+1| > 8$$

$$\therefore 3x+1 > 8 \quad \text{or} \quad 3x+1 < -8$$

$$\therefore 3x > 7$$

$$x > \frac{7}{3}$$

$$3x < -9$$

$$x < -3 \quad (2)$$

$$(f) m^{\frac{1}{x}} = n^3$$

$$\therefore \frac{1}{x} \log m = 3 \log n$$

$$\therefore \frac{1}{x} = \frac{3 \log n}{\log m} \quad (2)$$

$$\therefore x = \frac{\log m}{3 \log n} (= \frac{1}{3} \log_n m)$$

$$(g) m_1 = \frac{b}{4} \quad m_2 = -2$$

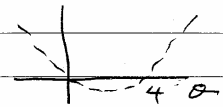
$$\therefore \frac{b}{4} = -2$$

$$\therefore b = -8 \quad (2)$$

$$(h) \theta^2 - 4\theta < 0$$

$$\theta(\theta-4) < 0$$

$$\therefore 0 < \theta < 4$$



$$(i) y = x^2 - 2x$$

$$y' = 2x - 2$$

$$\text{When } x=1 \quad y = -1$$

$$y' = 0$$

$$\therefore \text{eqn of tangent is } y = -1 \quad (2)$$

20

FOUR

$$c) 3x^2 + 4x + 5 \equiv A(x^2 + 2x + 1) + Bx + C$$

$$= Ax^2 + (2A+B)x + A+B+C$$

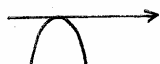
Hence $A=3$

$$2A+B=4 \Rightarrow B=-2$$

$$A+B+C=5 \Rightarrow C=4$$

$$b) P(x) = k - 3kx - x^2$$

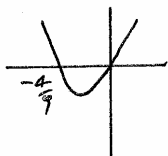
$$= -x^2 - 3kx + k$$

If $P(x) \leq 0$ then 

$$b^2 - 4ac \leq 0$$

$$9k^2 + 4k \leq 0$$

$$k(9k+4) \leq 0$$



$$-\frac{4}{9} \leq k \leq 0$$

$$c) x^4 - 10x^2 + 9 = 0$$

$$(x^2 - 1)(x^2 - 9) = 0$$

$$x^2 = 1, 9$$

$$x = \pm 1, \pm 3$$

$$d) y = x^2 + 6x + c$$

$$y' = 2x + 6$$

$$2x + 6 = 0 \text{ when } x = -3$$

$$b = -4$$

$$\text{Then } y = x^2 - 4x + c$$

Since $(2, 3)$ lies on curve

$$3 = 4 - 8 + c$$

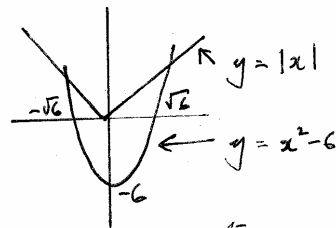
$$c = 7$$

$$e) x^2 - 4x + 4 + y^2 + 2y + 1 = 5$$

$$(x-2)^2 + (y+1)^2 = 5$$

Circle centre $(2, -1)$ radius $\sqrt{5}$

f)



Points of intersection

$$\text{When } y = x$$

$$x = x^2 - 6$$

$$x^2 - x - 6 = 0$$

$$(x-3)(x+2) = 0$$

$$x = 3 \text{ ONLY}$$

for right hand branch of $y = |x|$

$$\text{Hence } (3, 3)$$

When $y = -x$ (left hand branch)

$$-x = x^2 - 6$$

$$x^2 + x - 6 = 0$$

$$(x+3)(x-2) = 0$$

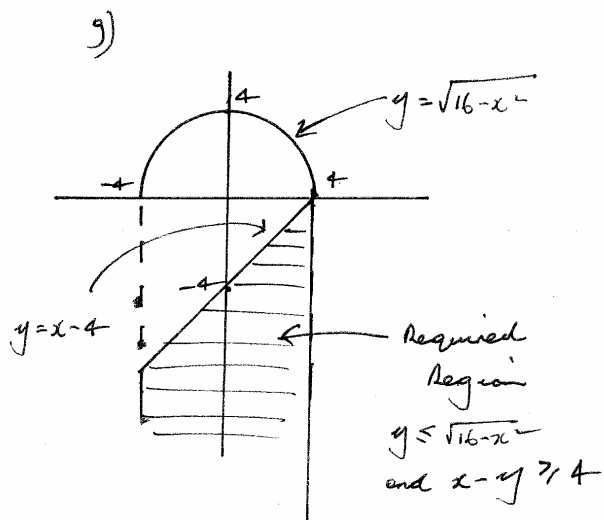
$$x = -3 \text{ ONLY}$$

$$\text{Hence } (-3, 3)$$

From sketch

$$x^2 - 6 \leq |x|$$

$$\text{When } -3 \leq x \leq 3$$



If $\cos A = \frac{4}{5}$

$$\tan A + \sec A = \frac{3}{4} + \frac{5}{4} = 2$$

$\cos A = \frac{4}{5}$ is a solution

If $\cos A = 0$
 $A = 90^\circ$

$\tan A + \sec A \neq 2$

Since $\tan 90^\circ$ is undefined

$\cos A = 0$ is not a solution.

h) $\tan A + \sec A = 2$

$$\frac{\sin A}{\cos A} + \frac{1}{\cos A} = 2$$

$$\frac{\sin A + 1}{\cos A} = 2$$

$$1 + \sin A = 2 \cos A$$

$$\sin A = 2 \cos A - 1$$

$$\sqrt{1 - \cos^2 A} = 2 \cos A - 1$$

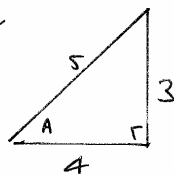
$$1 - \cos^2 A = 4 \cos^2 A - 4 \cos A + 1$$

$$5 \cos^2 A - 4 \cos A = 0$$

$$\cos A (5 \cos A - 4) = 0$$

Likely solutions are
 $\cos A = 0, \frac{4}{5}$

Check



QUESTIONS

(a) $f(x) = (1-4x) \times 2x + (x^2+5)x - 4$
 $= 2x - 8x^2 - 4x^2 - 20$
 $= -12x^2 + 2x - 20$

(b) Domain: $-\sqrt{3} < x < \sqrt{3}$ Range $y \geq \frac{1}{\sqrt{3}}$

(c) $h(-x) = -x^3 + x = -h(x)$
 $\therefore h(x)$ is an odd function

(d) as 5 cm is not the longest side, $\theta < 90^\circ$
 $5^2 = 6^2 + x^2 - 2 \times 6 \times x \cos \theta$ (Cos rule)
 $x^2 - 12x \cos \theta + 11 = 0$

Solution if $\Delta \geq 0$

$$\frac{12^2 \cos^2 \theta - 4 \times 11}{2} \geq 0$$

$$72 \cos^2 \theta - 22 \geq 0$$

$$\cos^2 \theta \geq \frac{11}{36}$$

$$\cos \theta \geq \frac{\sqrt{11}}{6} \text{ or } \cos \theta \leq -\frac{\sqrt{11}}{6}$$

$$\frac{\sqrt{11}}{6} \leq \cos \theta \quad \theta \text{ in 1st quadrant}$$

$$\cos \theta < 1 \text{ for all values of } \theta \text{ in 1st quadrant}$$

$$\therefore \frac{\sqrt{11}}{6} \leq \cos \theta < 1$$

(e) (i) In $\triangle ABC$ and $\triangle CDE$ $\angle ACB$ is common
 $\angle ABC = \angle EDC = 90^\circ$ (given)

$\therefore \triangle ABC$ is similar to $\triangle DEC$ (equiangular)

(ii) In $\triangle ABC$ and $\triangle CDE$ $\frac{BC}{DC} = \frac{AC}{CE}$ (Corresponding sides)

$$\therefore BC \times CE = AC \times CD$$

(iii) RHS = $AD \times DC - BE \times EC$
 $= (AC - DC)DC - (BC - EC)EC$
 $= AC \cdot CD - DC^2 - BC \cdot EC + EC^2$
 $= CE^2 - DC^2 + AC \cdot CD - BC \cdot EC$
 $= DE^2 \text{ as required } (AC \cdot CD = BC \cdot EC \text{ from (ii)})$

(f) $\alpha + \beta = -\frac{b}{a} = -5$, $\alpha\beta = \frac{c}{a} = 7$
 $(\alpha + \beta)^2 = 25$
 $(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$
 $= 25 - 28 = -3$

Required equation $[x - (\alpha - \beta)^2][x - (\alpha + \beta)^2]$
 $= (x + 3)(x - 25)$
 $= x^2 - 22x - 75$